Assignment 1: Deployment of an Image Classification REST API on Docker and VM

Task 1: Build and Test a Dockerized Image Classification REST API

1. **Create a Docker Image**
   • Package all necessary code, dependencies, and image classification model files into a Docker image.
   • Ensure the image serves the model via a REST API (e.g., using Flask or FastAPI).
2. **Run the Docker Container Locally**
   • Launch the container on your local machine.
   • Verify the API endpoints and model inference.
3. **Develop a Client Application**
   • Create a client app that sends images to the API.
   • Display classification results in a user-friendly format.

Task 2: Deploy the REST API on an Azure VM (Without Docker)

1. **Set Up the Azure VM**
   • Install required dependencies (Python, necessary libraries, etc.).
   • Configure networking to allow external access to the API.
2. **Deploy and Run the Model**
   • Transfer your model and API code to the VM.
   • Start the REST API service (e.g., using Gunicorn, Uvicorn, or Flask's built-in server).
3. **Access the API from Your Local Machine**
   • Test the deployed API by sending image requests from your local client app.
   • Validate response times and accuracy.

Task 3: Run the Voting app using Docker Compose

- Run the voting app as discussed in the lecture on your local computer using Docker Compose.
- Show running containers and UI interfaces for voting and result webapps.

Submission Guidelines

For each task, record a separate video demonstrating your solution, preferably through Loom. Submit the video URLs. Each video should be no longer than 3 minutes.

Video Requirements:

1. **Start with System Information**
   - Display your local computer's IP address and MAC address at the beginning.
2. **Demonstrate the Solution**
   - Run the solution step by step, showcasing its functionality.
   - Ensure clarity in execution, highlighting key aspects.
3. **Complete Workflow**
   - Show the end-to-end process, from setup to results.
   - Make sure the video effectively conveys how your solution works.

You are responsible for recording and submitting a clear, concise, and well-structured demonstration of your solution.